Prescribing Cost and Value SQL Project

Nov 2, 2026 · @Mustafa Malik

1. Perspective and End Consumer

This project looks at **prescribing cost and value** in primary care in England, using one month (August 2026) of the NHS Business Services Authority's English Prescribing Dataset. It is written for an **ICB medicines optimisation pharmacist**, who needs to know where prescribing money goes, which products and prescribers drive cost, and where there is scope to save without affecting patient care.

It is a Type 3 project: 15 questions, each built around a different family of SQL functions. The SQL is in `Prescribing_Queries.sql` and the database is built by `epd_setup.sql`.

Q	Function family	Main functions used
1	Aggregate	SUM, COUNT, COUNT(DISTINCT), AVG, MIN, MAX
2	Grouping	GROUP BY WITH ROLLUP, GROUPING()
3	String: extraction	LEFT, SUBSTRING_INDEX, TRIM
4	String: formatting	CONCAT, UPPER, LOWER, SUBSTRING, LOCATE
5	Pattern matching	REGEXP, LIKE
6	Numeric	ROUND, ABS, CEIL, FLOOR
7	Conditional	CASE, IF, NULLIF, COALESCE
8	Date	STR_TO_DATE, LAST_DAY, DAY, DATE_FORMAT
9	Ranking window	ROW_NUMBER, RANK
10	Distribution window	NTILE, PERCENT_RANK
11	Window aggregate	SUM() OVER with a running frame
12	Navigation window	LAG, FIRST_VALUE
13	Subqueries	Correlated subquery, EXISTS
14	Set operations	UNION ALL
15	Statistical aggregate	STDDEV_POP, VAR_POP

Conventions used throughout:

- **Cost** is the actual cost to the NHS after the national discount deduction; **NIC** (net ingredient cost) is the list price.
- **Items** are prescription items dispensed.
- BNF presentation codes for medicines (chapters 01–15) have 15 characters; a product code of **AA** at characters 10–11 marks a generic product.
- Practice-level comparisons use prescribers with **1,000+ items** in the month (2,000+ in Q13), so very small and non-GP settings do not distort averages.
- All figures are for **August 2026**.

2. Dataset

The English Prescribing Dataset lists every item prescribed in primary care in England and dispensed in the community, one row per prescriber, presentation and pack size. The August 2026 file has 17,754,211 rows (6.9 GB). It is loaded into the `prescribing` database as three tables.

Table	Rows	Contents
`prescriptions`	17,754,211	Month, practice code, presentation code, quantity, items, total quantity, ADQ usage, NIC, actual cost, unidentified flag, SNOMED code
`practices`	9,254	Practice name and postcode, sub-ICB, ICB (37) and NHS England region (7, plus one unidentified)
`presentations`	21,240	Presentation name, chemical substance (1,769 codes) and BNF chapter (20)

Loading

One load of all 27 columns would repeat every practice and drug name millions of times. Instead the file is read three times with `LOAD DATA LOCAL INFILE`: the first pass loads only codes and numbers into `prescriptions`, and the next two load `practices` and `presentations` with `IGNORE`, so the primary key keeps one row for each practice and presentation and discards the repeats. The file uses Windows line endings (`\r\n`) and writes numbers with a trailing full stop ("10."), which MySQL reads correctly. The full load takes about 20–40 minutes.

Data quality and limitations

Issue	How the project handles it
One month only	All figures are for August 2026; no trends over time
"Practices" include appliance, stoma, continence and other specialist services	Shown explicitly in Q4, Q10 and Q13; they inflate cost-per-item averages
One practice has an UNIDENTIFIED region and ICB	Excluded from regional questions
Chemical names (1,628) are fewer than chemical codes (1,769), because some names repeat across codes	Q8, Q9, Q11 and Q14 group by name
ADQ is a typical daily dose, not a patient count	Q8 figures are estimates of patients on treatment
Every row matched a practice and a presentation	No rows lost in the joins

3. Core Functions (Q1–Q8)

Aggregate, grouping, string, pattern-matching, numeric, conditional and date functions.

Q1: Aggregate functions

Question: What are the headline totals for August 2026: rows, practices, items, cost and NIC, and the average, lowest and highest cost per prescription line?

Concept: SUM, COUNT, COUNT(DISTINCT), AVG, MIN, MAX

prescription_lines	practices	presentations	items	actual_cost_gbp	nic_gbp	cost_per_item_gbp	avg_cost_per_line_gbp	min_line_cost_gbp	max_line_cost_gbp
17754211	9254	21240	102050994	904525400	943194934	8.86	50.95	0.00000	78561.19800

1 row returned (shown in full).

Interpretation: In August 2026, GP practices and other primary care prescribers in England issued 102.05 million items across 17.75 million prescription lines, at an actual cost of £904.5 million. That is £8.86 per item. List price (NIC) was £943.2 million, so actual cost was about £38.7 million lower after the national discount deduction. The most expensive single line cost £78,561.

Q2: Grouping functions

Question: How are items and cost split across NHS England regions and their ICBs, with a subtotal for each region and a grand total?

Concept: GROUP BY ... WITH ROLLUP + GROUPING() to label the subtotal and total rows

region	icb	items	cost_gbp_m	cost_per_item_gbp
EAST OF ENGLAND	NHS CENTRAL EAST INTEGRATED CARE BOARD	4967057	43.43	8.74
EAST OF ENGLAND	NHS ESSEX INTEGRATED CARE BOARD	3280385	29.50	8.99
EAST OF ENGLAND	NHS NORFOLK AND SUFFOLK INTEGRATED CARE BOARD	3623769	29.25	8.07
EAST OF ENGLAND	(region total)	11871211	102.18	8.61
LONDON	NHS NORTH EAST LONDON INTEGRATED CARE BOARD	3204020	26.54	8.28
LONDON	NHS SOUTH EAST LONDON INTEGRATED CARE BOARD	2260943	21.77	9.63
LONDON	NHS SOUTH WEST LONDON INTEGRATED CARE BOARD	2096129	18.07	8.62
LONDON	NHS WEST AND NORTH LONDON INTEGRATED CARE BOARD	5409778	46.47	8.59
LONDON	(region total)	12970870	112.85	8.70
MIDLANDS	NHS BIRMINGHAM AND SOLIHULL INTEGRATED CARE BOARD	2202170	23.90	10.85
MIDLANDS	NHS BLACK COUNTRY INTEGRATED CARE BOARD	2182596	22.81	10.45

region	icb	items	cost_gbp_m	cost_per_item_gbp
MIDLANDS	NHS COVENTRY AND WARWICKSHIRE INTEGRATED CARE BOARD	1779850	15.43	8.67
MIDLANDS	NHS DERBY AND DERBYSHIRE INTEGRATED CARE BOARD	1814946	15.68	8.64
MIDLANDS	NHS HEREFORDSHIRE AND WORCESTERSHIRE INTEGRATED CARE BOARD	1384773	14.21	10.26
MIDLANDS	NHS LEICESTER, LEICESTERSHIRE AND RUTLAND INTEGRATED CARE BOARD	2049358	19.20	9.37
MIDLANDS	NHS LINCOLNSHIRE INTEGRATED CARE BOARD	2048072	15.94	7.78
MIDLANDS	NHS NORTHAMPTONSHIRE INTEGRATED CARE BOARD	1451164	11.64	8.02
MIDLANDS	NHS NOTTINGHAM AND NOTTINGHAMSHIRE INTEGRATED CARE BOARD	2139729	17.32	8.10
MIDLANDS	NHS SHROPSHIRE, TELFORD AND WREKIN INTEGRATED CARE BOARD	862956	8.75	10.14
MIDLANDS	NHS STAFFORDSHIRE AND STOKE-ON-TRENT INTEGRATED CARE BOARD	2199068	19.89	9.04
MIDLANDS	(region total)	20114682	184.77	9.19

region	icb	items	cost_gbp_m	cost_per_item_gbp
NORTH EAST AND YORKSHIRE	NHS HUMBER AND NORTH YORKSHIRE INTEGRATED CARE BOARD	3663799	31.37	8.56
NORTH EAST AND YORKSHIRE	NHS NORTH EAST AND NORTH CUMBRIA INTEGRATED CARE BOARD	7445254	55.51	7.46
NORTH EAST AND YORKSHIRE	NHS SOUTH YORKSHIRE INTEGRATED CARE BOARD	3112186	23.65	7.60
NORTH EAST AND YORKSHIRE	NHS WEST YORKSHIRE INTEGRATED CARE BOARD	4670521	40.68	8.71
NORTH EAST AND YORKSHIRE	(region total)	18891760	151.21	8.00
NORTH WEST	NHS CHESHIRE AND MERSEYSIDE INTEGRATED CARE BOARD	5289397	46.07	8.71
NORTH WEST	NHS GREATER MANCHESTER INTEGRATED CARE BOARD	5571952	49.77	8.93
NORTH WEST	NHS LANCASHIRE AND SOUTH CUMBRIA INTEGRATED CARE BOARD	3449663	29.32	8.50
NORTH WEST	(region total)	14311012	125.16	8.75
SOUTH EAST	NHS HAMPSHIRE AND ISLE OF WIGHT INTEGRATED CARE BOARD	3155484	32.55	10.31

region	icb	items	cost_gbp_m	cost_per_item_gbp
SOUTH EAST	NHS KENT AND MEDWAY INTEGRATED CARE BOARD	3031141	30.01	9.90
SOUTH EAST	NHS SURREY AND SUSSEX INTEGRATED CARE BOARD	4519087	45.86	10.15
SOUTH EAST	NHS THAMES VALLEY INTEGRATED CARE BOARD	3117770	30.53	9.79
SOUTH EAST	(region total)	13823482	138.94	10.05
SOUTH WEST	NHS BATH AND NORTH EAST SOMERSET, SWINDON AND WILTSHIRE INTEGRATED CARE BOARD	1651115	14.34	8.68
SOUTH WEST	NHS BRISTOL, NORTH SOMERSET AND SOUTH GLOUCESTERSHIRE INTEGRATED CARE BOARD	1326481	13.24	9.98
SOUTH WEST	NHS CORNWALL AND THE ISLES OF SCILLY INTEGRATED CARE BOARD	1234358	10.77	8.73
SOUTH WEST	NHS DEVON INTEGRATED CARE BOARD	2304057	20.30	8.81
SOUTH WEST	NHS DORSET INTEGRATED CARE BOARD	1382572	11.72	8.48
SOUTH WEST	NHS GLOUCESTERSHIRE INTEGRATED CARE BOARD	1204135	9.71	8.06

region	icb	items	cost_gbp_m	cost_per_item_gbp
SOUTH WEST	NHS SOMERSET INTEGRATED CARE BOARD	952245	8.99	9.44
SOUTH WEST	(region total)	10054963	89.07	8.86
ALL REGIONS	(grand total)	102037980	904.17	8.86

44 rows returned (shown in full).

Interpretation: The Midlands spent the most (£184.8 million) and the South East had the highest cost per item (£10.05), while North East and Yorkshire had the lowest (£8.00). At ICB level, cost per item ranged from £7.46 in North East and North Cumbria to £10.85 in Birmingham and Solihull. The grand total (£904.2 million) is slightly below Q1 because the one unidentified region is excluded.

Q3: String functions: extraction

Question: Which BNF chapters account for the most spending? Split the chapter field ("20: Dressings") into its number and name.

Concept: CTE + LEFT, SUBSTRING_INDEX, TRIM + share of total

chapter_no	chapter_name	items	cost_gbp_m	pct_of_cost
06	Endocrine System	12126457	198.95	22.0
04	Central Nervous System	19413581	134.02	14.8
02	Cardiovascular System	31934760	109.84	12.1
03	Respiratory System	5695366	95.59	10.6
21	Appliances	3301979	76.99	8.5
09	Nutrition and Blood	5739359	70.10	7.8
01	Gastro-Intestinal System	9684155	54.99	6.1
23	Stoma Appliances	658439	35.54	3.9
07	Obstetrics, Gynaecology and Urinary-Tract Disorders	2930674	23.11	2.6
13	Skin	2035749	22.35	2.5

chapter_no	chapter_name	items	cost_gbp_m	pct_of_cost
05	Infections	2491241	16.75	1.9
08	Malignant Disease and Immunosuppression	377109	13.59	1.5
20	Dressings	401879	13.33	1.5
11	Eye	1242666	12.25	1.4
10	Musculoskeletal and Joint Diseases	2620271	11.00	1.2
12	Ear, Nose and Oropharynx	868619	6.07	0.7
22	Incontinence Appliances	225725	5.56	0.6
14	Immunological Products and Vaccines	121454	1.79	0.2
15	Anaesthesia	133106	1.68	0.2
19	Other Drugs and Preparations	48405	1.02	0.1

20 rows returned (shown in full).

Interpretation: The endocrine chapter accounts for 22.0% of spending (£199.0 million), more than any other, even though it has far fewer items than cardiovascular medicines. Cardiovascular has the most items (31.9 million) but only 12.1% of cost, because most of its medicines are cheap generics. Appliances, stoma and incontinence products and dressings (chapters 20–23) together make up 14.5% of spending.

Q4: String functions: formatting

Question: Which 15 practices had the highest prescribing cost? Show each name in sentence case with its postcode district and a single readable label.

Concept: CONCAT, UPPER, LOWER, LEFT, SUBSTRING, LOCATE

practice	postcode_district	label	items	cost_gbp
Thames valley appliance service	L3	Y08714 / S0E4D / south east	24063	1624280
Modality partnership (awc)	BD22	B83033 / QWO / north east and yorkshire	156116	1428166

practice	postcode_district	label	items	cost_gbp
Medicus health partners	N9	F85002 / Z9B2Z / london	141647	1142501
Nhs surrey icb - appliance service	L3	Y08792 / S9B9J / south east	14811	1111790
Bay medical group	LA3	Y01008 / QE1 / north west	105906	1017189
Affinity care	BD18	B83063 / QWO / north east and yorkshire	119022	1003810
Drs reddy and nunn	YO16	B81069 / QOQ / north east and yorkshire	111173	992284
Npc central northumberland	NE61	A84036 / QHM / north east and yorkshire	126154	895654
Portsdown group practice	PO2	J82155 / QRL / south east	95939	844422
Carlisle healthcare	CA1	A82016 / QHM / north east and yorkshire	71877	836751
Notts appliance management service	NG8	Y05430 / QT1 / midlands	9003	831301
Midlands medical partnership	B24	M85063 / QHL / midlands	81378	826830
Adhd assessment and treatment service	HA6	Y08511 / Z9B2Z / london	23967	826184
The willow group	PO12	J82083 / QRL / south east	95189	819973
Blmk appliance prescribing service	L3	Y07872 / S1Y5D / east of england	11518	817314

15 rows returned (shown in full). The label column uses " | " in the query output; it is shown with " / " here so the table renders.

Interpretation: Several of the highest-cost "practices" are not GP practices but appliance and specialist services. Thames Valley Appliance Service spent £1.62 million on just 24,063 items, and an ADHD assessment service appears too. The biggest true GP practice by cost is Modality Partnership (£1.43 million, 156,116 items). Simple sentence-casing lower-cases acronyms such as NHS and ICB, a limitation of using UPPER/LOWER without a lookup of exceptions.

Q5: Pattern matching

Question: For medicines (BNF chapters 01-15), what share of items and cost is generic rather than branded, and how many modified-release products are prescribed?

Concept: REGEXP on the BNF code structure + LIKE on the presentation name

product_type	items	pct_of_items	cost_gbp_m	pct_of_cost	cost_per_item_gbp	modified_release_items
Generic	84076791	86.3	410.78	53.2	4.89	4024373
Branded	13337776	13.7	361.30	46.8	27.09	340311

2 rows returned (shown in full).

Interpretation: Generic medicines make up 86.3% of items but only 53.2% of cost, at £4.89 per item against £27.09 for branded products, about 5.5 times more. Most modified-release prescribing is generic (4.0 million items against 0.3 million branded). Branded prescribing is where most of the remaining cost-saving opportunity lies, although many high-cost brands have no generic equivalent yet.

Q6: Numeric functions

Question: For each BNF chapter, how far is the actual cost below the list price (NIC), and what is the cost per item rounded up and down to the nearest pound?

Concept: ROUND, ABS, CEIL, FLOOR

chapter_no	chapter_name	nic_gbp	actual_cost_gbp	difference_gbp	pct_below_nic	cost_per_item_gbp	cost_per_item_floor	cost_per_item_ceil
06	Endocrine System	204612951	198953411	5659540	2.77	16.41	16	17
04	Central Nervous System	140132382	134019633	6112749	4.36	6.90	6	7
02	Cardiovascular System	115912724	109837375	6075348	5.24	3.44	3	4
03	Respiratory System	99387871	95587332	3800540	3.82	16.78	16	17
21	Appliances	79230626	76987547	2243079	2.83	23.32	23	24
09	Nutrition and Blood	71843937	70102667	1741270	2.42	12.21	12	13

chapter_no	chapter_name	nic_gbp	actual_cost_gbp	difference_gbp	pct_below_nic	cost_per_item_gbp	cost_per_item_floor	cost_per_item_ceil
01	Gastro-Intestinal System	58896915	54994958	3901957	6.63	5.68	5	6
23	Stoma Appliances	35767420	35536946	230474	0.64	53.97	53	54
07	Obstetrics, Gynaecology and Urinary-Tract Disorders	25143160	23108964	2034195	8.09	7.89	7	8
13	Skin	23760978	22347677	1413301	5.95	10.98	10	11
05	Infections	18340516	16754054	1586462	8.65	6.73	6	7
08	Malignant Disease and Immunosuppression	14179124	13591757	587367	4.14	36.04	36	37
20	Dressings	13994282	13326561	667722	4.77	33.16	33	34
11	Eye	13108672	12250801	857871	6.54	9.86	9	10
10	Musculoskeletal and Joint Diseases	11971528	11000800	970728	8.11	4.20	4	5
12	Ear, Nose and Oropharynx	6539090	6070201	468889	7.17	6.99	6	7
22	Incontinence Appliances	5629025	5564867	64159	1.14	24.65	24	25
14	Immunological Products and Vaccines	1915261	1785230	130030	6.79	14.70	14	15
15	Anaesthesia	1773965	1680937	93028	5.24	12.63	12	13
19	Other Drugs and Preparations	1054507	1023682	30825	2.92	21.15	21	22

20 rows returned (shown in full).

Interpretation: Actual cost is below list price in every chapter, by between 0.64% (stoma appliances) and 8.65% (infections). Medicines chapters with many generics, such as infections, musculoskeletal and obstetrics, have the largest deductions (8–9%), while appliances and endocrine products have the smallest. Stoma appliances also have the highest cost per item (£53.97, rounding down to £53 and up to £54).

Q7: Conditional functions

Question: Group practices by the number of items they dispensed in the month. How do cost per item and the share of unidentified prescribing differ by practice size?

Concept: CASE, IF, NULLIF (safe division), COALESCE

size_band	practices	items	cost_per_item_gbp	pct_unidentified
1. Under 1,000	2745	420177	17.06	2.87
2. 1,000-4,999	804	2337146	12.21	0.31
3. 5,000-9,999	1462	11147317	9.10	0.00
4. 10,000-19,999	2590	37514179	8.95	0.03

size_band	practices	items	cost_per_item_gbp	pct_unidentified
5. 20,000+	1653	50632175	8.52	0.00

5 rows returned (shown in full).

Interpretation: The 2,745 prescribers with under 1,000 items are mostly specialist services rather than GP practices, and they have the highest cost per item (£17.06) and the most unidentified prescribing (2.87%). Cost per item falls steadily with size, to £8.52 in practices with 20,000+ items, and unidentified prescribing is almost zero in larger practices.

Q8: Date functions

Question: Turn the month field into a real date. Using the number of days in the month, how many patients could the 15 most-used medicines treat each day, based on average daily quantities (ADQ)?

Concept: STR_TO_DATE, LAST_DAY, DAY, DATE_FORMAT

chemical_name	month	days_in_month	items	total_adq	patients_treated_per_day
Atorvastatin	August 2026	31	6447607	335562623	10824601
Amlodipine	August 2026	31	3300882	153869894	4963545
Lansoprazole	August 2026	31	3053376	112394394	3625626
Ramipril	August 2026	31	2925802	108035195	3485006
Omeprazole	August 2026	31	2737893	101164705	3263378
Sertraline hydrochloride	August 2026	31	2110491	100763341	3250430
Metformin hydrochloride	August 2026	31	2222751	68920003	2223226
Budesonide	August 2026	31	422536	63639539	2052888
Salbutamol	August 2026	31	1037738	61085864	1970512
Levothyroxine sodium	August 2026	31	2802742	60233767	1943025
Losartan potassium	August 2026	31	1149142	48185648	1554376

chemical_name	month	days_in_month	items	total_adq	patients_treated_per_day
Candesartan cilexetil	August 2026	31	747624	36283746	1170443
Fexofenadine hydrochloride	August 2026	31	630845	34344576	1107890
Citalopram hydrobromide	August 2026	31	988670	31634347	1020463
Fluoxetine hydrochloride	August 2026	31	604840	28159890	908384

15 rows returned (shown in full).

Interpretation: Atorvastatin's average daily quantities are enough to treat about 10.8 million people every day in August, more than twice any other medicine. Amlodipine (5.0 million), lansoprazole, ramipril and omeprazole (3.3–3.6 million each) follow. ADQ is a typical maintenance dose, so these figures are estimates of patients on treatment, not exact counts.

4. Window Functions (Q9–Q12)

Ranking, distribution, running-total and navigation window functions.

Q9: Ranking window functions

Question: What are the three most expensive chemical substances in each NHS England region?

Concept: ROW_NUMBER and RANK partitioned by region

region_name	cost_rank	chemical_name	cost_gbp_m	items	items_rank
EAST OF ENGLAND	1	Tirzepatide	9.33	40982	67
EAST OF ENGLAND	2	Detection Sensor Interstitial Fluid/Gluc	3.81	49749	52
EAST OF ENGLAND	3	BeclometDiprop/Formoterol	2.27	87054	30
LONDON	1	Tirzepatide	10.04	38774	78
LONDON	2	Detection Sensor Interstitial Fluid/Gluc	4.04	44535	64
LONDON	3	BeclometDiprop/Formoterol	2.56	77796	35
MIDLANDS	1	Tirzepatide	16.60	71452	68
MIDLANDS	2	Detection Sensor Interstitial Fluid/Gluc	6.21	77294	61

region_name	cost_rank	chemical_name	cost_gbp_m	items	items_rank
MIDLANDS	3	BeclometDiprop/Formoterol	4.97	170035	27
NORTH EAST AND YORKSHIRE	1	Tirzepatide	12.44	57846	75
NORTH EAST AND YORKSHIRE	2	Detection Sensor Interstitial Fluid/Gluc	5.40	65650	69
NORTH EAST AND YORKSHIRE	3	BeclometDiprop/Formoterol	3.54	121580	37
NORTH WEST	1	Tirzepatide	9.02	39464	86
NORTH WEST	2	Detection Sensor Interstitial Fluid/Gluc	4.44	57695	56
NORTH WEST	3	BeclometDiprop/Formoterol	3.09	114255	25
SOUTH EAST	1	Tirzepatide	14.29	59330	54
SOUTH EAST	2	Detection Sensor Interstitial Fluid/Gluc	5.93	67382	46
SOUTH EAST	3	BeclometDiprop/Formoterol	3.13	93996	31
SOUTH WEST	1	Tirzepatide	8.11	34358	73
SOUTH WEST	2	Detection Sensor Interstitial Fluid/Gluc	3.48	42501	56
SOUTH WEST	3	BeclometDiprop/Formoterol	2.11	71240	31

21 rows returned (shown in full).

Interpretation: The same three products top spending in every region: tirzepatide, glucose monitoring sensors and the beclometasone/formoterol inhaler. Tirzepatide costs £8–17 million per region yet ranks only 54th to 86th by items, showing how a single high-cost new medicine now dominates prescribing budgets.

Q10: Distribution window functions

Question: Split practices (1,000+ items) into ten equal groups by cost per item. What range does each decile cover, and how many practices are in the most expensive 5%?

Concept: NTILE and PERCENT_RANK

decile	practices	min_cost_per_item	max_cost_per_item	avg_cost_per_item	in_top_5_pct
1	651	1.64	6.69	5.73	0
2	651	6.69	7.31	7.03	0
3	651	7.31	7.74	7.53	0
4	651	7.74	8.17	7.96	0

decile	practices	min_cost_per_item	max_cost_per_item	avg_cost_per_item	in_top_5_pct
5	651	8.17	8.58	8.38	0
6	651	8.58	9.02	8.80	0
7	651	9.02	9.59	9.29	0
8	651	9.59	10.29	9.91	0
9	651	10.29	11.42	10.80	0
10	650	11.42	119.00	19.10	326

10 rows returned (shown in full).

Interpretation: For practices with 1,000+ items, the first nine deciles span a narrow band of £1.64 to £11.42 per item, but the top decile stretches to £119.00 and averages £19.10. That top tail, including the 326 practices in the top 5%, is largely made up of appliance, stoma and specialist services (see Q13), which should be benchmarked separately from GP practices.

Q11: Window aggregate functions

Question: How concentrated is spending? Rank chemical substances by cost and use a running total to find how many account for 50%, 80%, 90% and 99% of all spending.

Concept: SUM() OVER (ORDER BY ...) running total + share of total

threshold_pct	chemicals_needed	total_chemicals	pct_of_chemicals
50	43	1628	2.6
80	179	1628	11.0
90	311	1628	19.1
99	762	1628	46.8

4 rows returned (shown in full).

Interpretation: Spending is highly concentrated: just 43 of 1,628 chemical substances (2.6%) account for half of all spending, and 179 (11.0%) for 80%. Even 99% of spending is covered by fewer than half of all chemicals (762). Medicines optimisation work focused on a few dozen products can therefore reach most of the budget.

Q12: Navigation window functions

Question: In each of the four most expensive BNF chapters, what are the top five chemicals by cost, how far is each behind the one above it, and what share of the chapter leader's cost does it reach?

Concept: LAG and FIRST_VALUE partitioned by chapter

bnf_chapter	chapter_rank	chemical_name	cost_gbp_m	gap_to_above_gbp_m	pct_of_leader
02: Cardiovascular System	1	Sacubitril/valsartan	14.23	NULL	100.0
02: Cardiovascular System	2	Ramipril	12.59	1.64	88.4
02: Cardiovascular System	3	Edoxaban	12.31	0.28	86.5
02: Cardiovascular System	4	Atorvastatin	5.94	6.37	41.8
02: Cardiovascular System	5	Bisoprolol fumarate	4.64	1.30	32.6
03: Respiratory System	1	BeclometDiprop/Formoterol	21.67	NULL	100.0
03: Respiratory System	2	BeclometDiprop/Formoterol/Glycopyrronium (Corticosteroids)	15.50	6.17	71.5
03: Respiratory System	3	Budesonide	13.30	2.20	61.4
03: Respiratory System	4	Fluticasone/umeclidinium/vilanterol	6.48	6.83	29.9
03: Respiratory System	5	Adrenaline	5.65	0.83	26.1
04: Central Nervous System	1	Lisdexamfetamine dimesylate	8.80	NULL	100.0
04: Central Nervous System	2	Co-codamol (Codeine phosphate/paracetamol)	6.47	2.34	73.5
04: Central Nervous System	3	Methylphenidate hydrochloride	6.02	0.45	68.3
04: Central Nervous System	4	Paracetamol	5.72	0.30	64.9
04: Central Nervous System	5	Sumatriptan succinate	5.09	0.62	57.8
06: Endocrine System	1	Tirzepatide	79.83	NULL	100.0
06: Endocrine System	2	Semaglutide	15.06	64.78	18.9
06: Endocrine System	3	Estradiol	9.35	5.70	11.7
06: Endocrine System	4	Insulin aspart	9.14	0.21	11.5
06: Endocrine System	5	Empagliflozin	8.95	0.19	11.2

20 rows returned (shown in full).

Interpretation: Tirzepatide dominates the endocrine chapter at £79.8 million, more than five times the next product, semaglutide (18.9% of the leader). In the other chapters the top products are closer together: sacubitril/valsartan, ramipril and edoxaban are within 14% of each other in cardiovascular, and lisdexamfetamine (an ADHD medicine) leads the CNS chapter at £8.8 million.

5. Subqueries, Set Operations and Statistics (Q13–Q15)

Correlated subqueries with EXISTS, set operations and statistical aggregates.

Q13: Subqueries

Question: Which practices (2,000+ items) have a cost per item more than 50% above the average for their own ICB, and do they prescribe any of the ten most expensive chemicals nationally? Show the 20 furthest above.

Concept: Correlated subquery (ICB average per row) + EXISTS against a subquery of top chemicals

practice_name	icb_name	items	cost_per_item_gbp	icb_avg_cost_per_item	prescribes_top10_chemical
NOTTS APPLIANCE MANAGEMENT SERVICE	NHS NOTTINGHAM AND NOTTINGHAMSHIRE INTEGRATED CARE BOARD	9003	92.34	8.85	Yes
MARSS NHS LIVERPOOL CCG	NHS CHESHIRE AND MERSEYSIDE INTEGRATED CARE BOARD	3402	87.34	9.43	No
MARSS NHS WIRRAL CCG	NHS CHESHIRE AND MERSEYSIDE INTEGRATED CARE BOARD	3613	85.02	9.43	Yes
SNAPS STOMA	NHS CHESHIRE AND MERSEYSIDE INTEGRATED CARE BOARD	2251	79.39	9.43	Yes
STOCKPORT CONTINENCE AND STOMA SERVICE	NHS GREATER MANCHESTER INTEGRATED CARE BOARD	4188	77.15	9.20	Yes
SSCCG CONTINENCE & STOMA SERVICE	NHS CHESHIRE AND MERSEYSIDE INTEGRATED CARE BOARD	4750	77.71	9.43	Yes
LANCASTER HOUSE CONTINENCE	NHS GREATER MANCHESTER INTEGRATED CARE BOARD	2590	75.01	9.20	Yes
APPLIANCE PRESCRIPTION SERVICE	NHS LANCASHIRE AND SOUTH CUMBRIA INTEGRATED CARE BOARD	7804	72.31	8.96	Yes

practice_name	icb_name	items	cost_per_item_gbp	icb_avg_cost_per_item	prescribes_top10_chemical
BLMK APPLIANCE PRESCRIBING SERVICE	NHS CENTRAL EAST INTEGRATED CARE BOARD	11518	70.96	8.92	Yes
SOUTH COUNTY CONTINENCE PRESCRIPTION SER	NHS NOTTINGHAM AND NOTTINGHAMSHIRE INTEGRATED CARE BOARD	2171	69.71	8.85	Yes
UHBW STOMA PRESCRIBING SERVICE	NHS BRISTOL, NORTH SOMERSET AND SOUTH GLOUCESTERSHIRE INTEGRATED CARE BOARD	2239	93.19	11.83	No
NHS SURREY ICB - APPLIANCE SERVICE	NHS SURREY AND SUSSEX INTEGRATED CARE BOARD	14811	75.07	10.33	Yes
SNEE APMS	NHS NORFOLK AND SUFFOLK INTEGRATED CARE BOARD	7024	58.82	8.27	Yes
THAMES VALLEY APPLIANCE SERVICE	NHS THAMES VALLEY INTEGRATED CARE BOARD	24063	67.50	9.68	Yes
SOUTH EAST HANTS CONTINENCE SERVICE	NHS HAMPSHIRE AND ISLE OF WIGHT INTEGRATED CARE BOARD	2419	81.07	11.74	Yes
LLR APPLIANCE PRESCRIPTION SERVICE	NHS LEICESTER, LEICESTERSHIRE AND RUTLAND INTEGRATED CARE BOARD	6017	62.78	9.59	Yes
FAREHAM & GOSPORT CONTINENCE SERVICE	NHS HAMPSHIRE AND ISLE OF WIGHT INTEGRATED CARE BOARD	2188	76.24	11.74	Yes
STOMA PRESCRIBING SERVICE (NBT)	NHS BRISTOL, NORTH SOMERSET AND SOUTH GLOUCESTERSHIRE INTEGRATED CARE BOARD	2503	73.72	11.83	Yes
NORTH & MID CONTINENCE SERVICE	NHS HAMPSHIRE AND ISLE OF WIGHT INTEGRATED CARE BOARD	2055	69.33	11.74	Yes

practice_name	icb_name	items	cost_per_item_gbp	icb_avg_cost_per_item	prescribes_top10_chemical
CATHETER PRESCRIPTION SERVICE	NHS CHESHIRE AND MERSEYSIDE INTEGRATED CARE BOARD	3994	52.39	9.43	Yes

20 rows returned (shown in full).

Interpretation: All 20 prescribers furthest above their ICB average are appliance, stoma, continence or catheter services, not GP practices. Their cost per item is £52–93, against ICB averages of about £8–12. 18 of the 20 prescribe one of the ten most expensive chemicals nationally, mostly catheters and stoma bags. Benchmarking GP practices should exclude these services.

Q14: Set operations

Question: Compare the 20 most-prescribed chemicals by items with the 20 most expensive by cost. Which appear in both lists, and which in only one?

Concept: UNION ALL of two ranked lists + GROUP BY to classify

chemical_name	appears_in	items	cost_gbp_m	cost_per_item_gbp
Ramipril	Both lists	2925802	12.59	4.30
Colecalciferol	Both lists	2350864	9.12	3.88
Omeprazole	Both lists	2737898	7.54	2.75
Tirzepatide	High cost only	342252	79.83	233.26
Detection Sensor Interstitial Fluid/Gluc	High cost only	404817	33.31	82.29
BeclometDiprop/Formoterol	High cost only	735983	21.67	29.44
Catheters	High cost only	124835	16.34	130.93
BeclometDiprop/Formoterol/Glycopyrronium (Corticosteroids)	High cost only	316563	15.50	48.97
Semaglutide	High cost only	144757	15.06	104.02
Sacubitril/valsartan	High cost only	168743	14.23	84.33
Budesonide	High cost only	426634	14.03	32.88
Edoxaban	High cost only	267603	12.31	46.01
Estradiol	High cost only	766874	11.56	15.07
Ileostomy Bags	High cost only	65708	11.33	172.46
Wound Management & Other Dressings	High cost only	276357	10.34	37.41
Insulin aspart	High cost only	220401	9.14	41.49
Empagliflozin	High cost only	250482	8.95	35.73
Lisdexamfetamine dimesylate	High cost only	135983	8.80	64.74

chemical_name	appears_in	items	cost_gbp_m	cost_per_item_gbp
Colostomy Bags	High cost only	36800	8.76	237.97
Insulin glargine	High cost only	177210	7.68	43.33
Co-codamol (Codeine phosphate/paracetamol)	High volume only	1137733	6.47	5.68
Atorvastatin	High volume only	6447607	5.94	0.92
Paracetamol	High volume only	1225534	5.72	4.66
Metformin hydrochloride	High volume only	2222751	5.40	2.43
Bisoprolol fumarate	High volume only	2550397	4.64	1.82
Aspirin	High volume only	1623176	4.32	2.66
Sertraline hydrochloride	High volume only	2110491	4.13	1.96
Dapagliflozin	High volume only	1288442	3.79	2.94
Salbutamol	High volume only	1037746	3.35	3.23
Lansoprazole	High volume only	3053376	3.23	1.06
Levothyroxine sodium	High volume only	2802814	3.20	1.14
Amlodipine	High volume only	3301130	2.66	0.81
Mirtazapine	High volume only	1119388	1.57	1.40
Amitriptyline hydrochloride	High volume only	1317568	1.22	0.93
Citalopram hydrobromide	High volume only	993195	0.97	0.98
Losartan potassium	High volume only	1149167	0.95	0.83
Clopidogrel	High volume only	996820	0.83	0.83

37 rows returned (shown in full).

Interpretation: Only three chemicals are both high volume and high cost: ramipril, colecalciferol and omeprazole. The 17 high-cost-only products are newer or specialist items, led by tirzepatide at £233 per item, while the 17 high-volume-only products are cheap generics such as atorvastatin, which has 6.4 million items at just £0.92 each. Volume and cost need separate strategies.

Q15: Statistical aggregate functions

Question: How much do practices within each ICB vary in cost per item? Calculate the mean, standard deviation, variance and coefficient of variation for practices with 1,000+ items.

Concept: AVG, STDDEV_POP, VAR_POP + coefficient of variation

icb_name	practices	mean_cost_per_item	sd	variance	cv_pct
NHS SOUTH YORKSHIRE INTEGRATED CARE BOARD	186	9.04	10.56	111.47	116.8
NHS CHESHIRE AND MERSEYSIDE INTEGRATED CARE BOARD	359	10.44	12.14	147.47	116.3
NHS NOTTINGHAM AND NOTTINGHAMSHIRE INTEGRATED CARE BOARD	132	9.50	10.47	109.58	110.1
NHS BRISTOL, NORTH SOMERSET AND SOUTH GLOUCESTERSHIRE INTEGRATED CARE BOARD	73	11.83	12.23	149.66	103.4
NHS GREATER MANCHESTER INTEGRATED CARE BOARD	435	9.91	9.55	91.28	96.5
NHS HAMPSHIRE AND ISLE OF WIGHT INTEGRATED CARE BOARD	156	12.36	11.63	135.33	94.1
NHS NORFOLK AND SUFFOLK INTEGRATED CARE BOARD	167	8.64	6.79	46.07	78.5
NHS LINCOLNSHIRE INTEGRATED CARE BOARD	85	8.62	6.54	42.83	75.9
NHS BIRMINGHAM AND SOLIHULL INTEGRATED CARE BOARD	189	12.00	8.34	69.55	69.5
NHS SHROPSHIRE, TELFORD AND WREKIN INTEGRATED CARE BOARD	54	11.10	7.5	56.22	67.5
NHS CORNWALL AND THE ISLES OF SCILLY INTEGRATED CARE BOARD	59	9.52	6.31	39.84	66.3
NHS ESSEX INTEGRATED CARE BOARD	217	9.88	6.5	42.22	65.7
NHS NORTH EAST AND NORTH CUMBRIA INTEGRATED CARE BOARD	372	7.85	5.1	26.03	65
NHS LANCASHIRE AND SOUTH CUMBRIA INTEGRATED CARE BOARD	208	9.06	5.43	29.43	59.9
NHS LEICESTER, LEICESTERSHIRE AND RUTLAND INTEGRATED CARE BOARD	132	9.63	5.57	30.99	57.8
NHS BATH AND NORTH EAST SOMERSET, SWINDON AND WILTSHIRE INTEGRATED CARE BOARD	87	9.17	4.85	23.48	52.9
NHS CENTRAL EAST INTEGRATED CARE BOARD	277	8.97	4.67	21.82	52.1
NHS HEREFORDSHIRE AND WORCESTERSHIRE INTEGRATED CARE BOARD	81	10.94	5.35	28.6	48.9

icb_name	practices	mean_cost_per_item	sd	variance	cv_pct
NHS THAMES VALLEY INTEGRATED CARE BOARD	206	9.50	4.54	20.58	47.8
NHS SURREY AND SUSSEX INTEGRATED CARE BOARD	303	10.74	4.88	23.86	45.5
NHS SOUTH WEST LONDON INTEGRATED CARE BOARD	174	8.91	3.81	14.51	42.8
NHS DEVON INTEGRATED CARE BOARD	119	8.99	3.47	12.06	38.6
NHS HUMBER AND NORTH YORKSHIRE INTEGRATED CARE BOARD	162	8.88	3.24	10.47	36.4
NHS WEST AND NORTH LONDON INTEGRATED CARE BOARD	527	8.65	2.96	8.76	34.2
NHS KENT AND MEDWAY INTEGRATED CARE BOARD	182	10.00	3.29	10.8	32.9
NHS BLACK COUNTRY INTEGRATED CARE BOARD	182	10.52	3.39	11.49	32.2
NHS SOUTH EAST LONDON INTEGRATED CARE BOARD	194	9.72	2.63	6.93	27.1
NHS WEST YORKSHIRE INTEGRATED CARE BOARD	274	8.73	2.37	5.61	27.1
NHS DERBY AND DERBYSHIRE INTEGRATED CARE BOARD	115	8.61	2.12	4.51	24.7
NHS NORTH EAST LONDON INTEGRATED CARE BOARD	283	8.50	2.08	4.32	24.4
NHS STAFFORDSHIRE AND STOKE-ON-TRENT INTEGRATED CARE BOARD	144	9.03	2.1	4.41	23.3
NHS SOMERSET INTEGRATED CARE BOARD	57	9.25	1.69	2.86	18.3
NHS COVENTRY AND WARWICKSHIRE INTEGRATED CARE BOARD	120	8.75	1.59	2.52	18.2
NHS DORSET INTEGRATED CARE BOARD	68	8.52	1.38	1.9	16.2
NHS GLOUCESTERSHIRE INTEGRATED CARE BOARD	64	8.10	1.31	1.71	16.1
NHS NORTHAMPTONSHIRE INTEGRATED CARE BOARD	65	7.85	0.84	0.7	10.7
UNIDENTIFIED	1	27.14	0	0	0

37 rows returned (shown in full).

Interpretation: Variation between practices within an ICB ranges from a coefficient of variation of 10.7% in Northamptonshire to 116.8% in South Yorkshire. The ICBs with the highest variation, such as Cheshire and Merseyside, include appliance and continence services among their prescribers (see Q13), which inflates the spread. ICBs with low variation have more uniform GP prescribing. The single unidentified prescriber can be ignored.

6. Summary of Findings

For an ICB medicines optimisation pharmacist, the August 2026 data points to five priorities:

1. **A few products drive most of the spending.** Just 43 of 1,628 chemicals (2.6%) account for half of the £904.5 million, and 179 account for 80% (Q11). Tirzepatide alone cost £79.8 million, is the top-cost product in every region and costs £233 per item (Q9, Q12, Q14). Reviewing these few products will reach most of the budget.
2. **Endocrine is now the most expensive chapter.** It accounts for 22.0% of cost (£199.0 million), well ahead of cardiovascular, which has the most items but only 12.1% of cost (Q3). Weight-management and diabetes medicines are the main pressure.
3. **Branded prescribing is where the unit-cost gap is.** Generics make up 86.3% of medicine items but only 53.2% of cost, at £4.89 per item against £27.09 for brands (Q5). High-volume generics such as atorvastatin cost under £1 per item (Q14).
4. **Benchmark GP practices separately from appliance services.** Appliance, stoma, continence and catheter services fill the top of every cost-per-item ranking (Q4, Q10, Q13) at £52–93 per item, and they inflate the variation in ICBs such as Cheshire and Merseyside and South Yorkshire (Q15). Comparisons are only fair once these are separated out.
5. **Regional and ICB variation is real but moderate.** Cost per item runs from £8.00 (North East and Yorkshire) to £10.05 (South East) by region, and from £7.46 to £10.85 by ICB (Q2). Within-ICB variation ranges from a coefficient of variation of 10.7% to 116.8% (Q15).

Limitations: This is a single month, so seasonal patterns and trends cannot be seen. Practice codes include non-GP prescribers. ADQ-based patient estimates (Q8) are approximate, and cost per item does not adjust for practice list size or patient need.